MuRoDance: Music-Anticipatory Humanoid Dance through Discrete–Continuous Motion Generation

Abstract— **Humanoid dance to arriving music requires anticipating musical events while continuing coherently from motion already committed. We present ForeDance, a framework for streaming music-driven humanoid dance that combines predicted musical context with robot-native discrete–continuous (D+C) motion generation. A frozen music foundation model forecasts future music states from heard audio alone. These states are aligned with upcoming motion times and condition both discrete structural planning and continuous residual generation. Kinematics-guided reconstruction objectives assign explicit structural responsibilities to discrete codes, while their combination with continuous residuals reconstructs full motion through a shared decoder. Commit Forcing trains continuation from jointly committed structure and detail and the reference boundary state reconstructed from their decoded motion. Committed references are delivered to a fixed whole-body controller, whose measured-state feedback remains separate from the generator's reference-context update. On the evaluated 60-second FineDance generations, a frozen music–motion evaluator favors the intended soundtrack over alternative tracks in approximately 75% of pairwise comparisons under the reported evaluation and selection protocol. Matched history-training comparisons reveal a tradeoff between motion distribution fidelity and smoothness. These results characterize musical correspondence and motion quality in generated robot-native references.**

I. Introduction

Dance preparation begins before a visible accent: weight shifts precede a step, and the body transitions toward a pose before reaching it. A humanoid dancing to arriving music must prepare its movements before hearing all the sound they will accompany. Waiting for an event leaves less time to prepare; reading the future recording changes the task into offline generation. Streaming further requires each new movement to continue from emitted motion that can no longer be revised. Musical anticipation and continuation from committed motion are therefore coupled challenges for music-driven humanoid dance.

Music-conditioned choreography has advanced through expressive motion models and large paired datasets [1]–[4]. DiscoForcing [5] extends causal audio-driven generation to streaming motion and humanoid deployment, while RoboPerform [6] connects audio-derived style with expressive humanoid control. Building on these advances, we study how a humanoid can continuously generate dance that is appropriate to arriving music. Music does not prescribe a unique choreography, but its rhythm, accents, and dynamic variation should be reflected in coherent movement and remain perceptible after execution. This motivates a coupled question: how can predicted musical context guide upcoming robot-native motion while preserving continuity with the movement already committed?

We present ForeDance, a framework for streaming music-driven humanoid dance that uses predicted musical context to guide upcoming movement. A frozen Magenta RealTime 2 (MRT-2) music foundation model [7] forecasts future music states from the audio received so far (Fig. 1(a)), providing musical lookahead without access to future waveform samples. Music pretraining supplies the continuation prior, while paired dance data teaches the robot-native motion generator to interpret its predictions. Aligned with upcoming motion times, these forecast states condition both discrete structural planning and continuous-detail generation, connecting musical anticipation to whole-body organization and local articulation.

Expressing musical conditions through a humanoid requires coordinated whole-body structure, expressive local articulation, and continuity across successive motion decisions. Our kinematics-guided discrete–continuous (D+C) representation couples discrete structure and continuous detail through complementary reconstruction objectives (Fig. 1(c)). Structure-only reconstruction supervises designated body organization and endpoint placement, while the combined latent reconstructs full motion through a shared boundary-conditioned decoder. Building on hybrid motion representations [8], this supervision assigns explicit kinematic responsibilities to the discrete component while retaining complementary variation in the full latent.

During streaming generation, both the motion history and the next boundary state depend on the model's own output rather than recorded motion. Commit Forcing trains continuation after jointly sampling and committing both components (Fig. 1(d)). Their decoded movement determines the next reference boundary state, pairing generated history with the state it actually induces. The next plan is trained using this generated history–state pair.

At each decision, ForeDance plans a short motion segment, commits a prefix, and updates its reference context for the next plan (Fig. 1(b)). A reference interface delivers the committed motion to a fixed SONIC whole-body controller [9]. The generator continues from reference-derived history and boundary state, while measured robot state closes the controller's physical feedback loop. Because reference delivery and tracking can alter timing and motion dynamics, we formulate a paired evaluation protocol to examine music–motion matching, music adaptation, and dance quality in generated references and executed trajectories.

Our contributions are:

- **Anticipatory music conditioning for humanoid dance.**

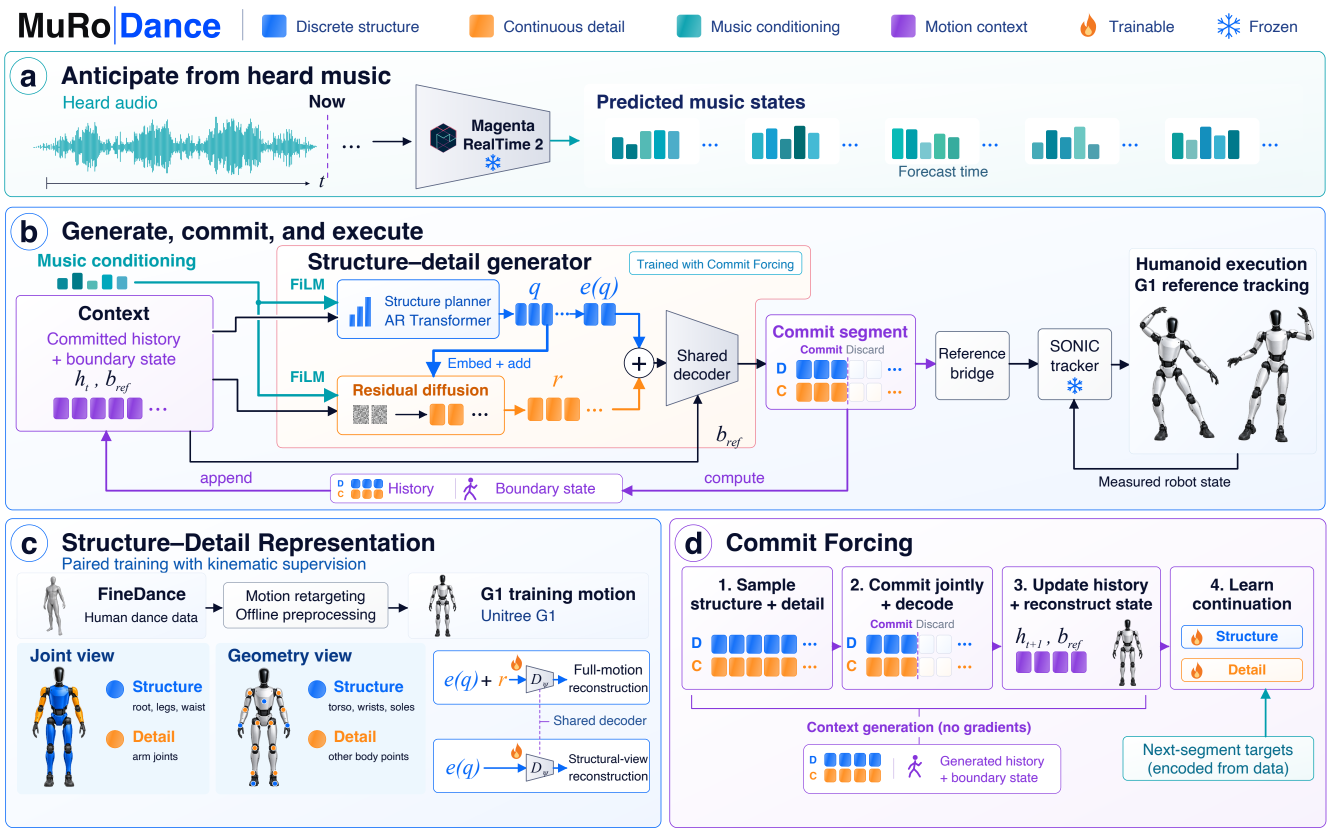


Fig. 1. **Overview of ForeDance. (a)** Frozen MRT-2 forecasts upcoming music states from arrived audio. **(b)** Forecast conditions modulate the discrete planner and residual diffusion model through FiLM; sampled structure also conditions residual generation. A shared decoder produces robot-native references, whose committed prefix is delivered through a bridge to fixed SONIC control. Committed latent codes update the generator history, and decoded motion determines its next reference boundary state. Measured robot state feeds back only to the controller. **(c)** Retargeted FineDance motion supports D+C learning through full-motion and structural-view reconstruction with a shared decoder and complementary kinematic supervision. **(d)** Commit Forcing constructs a generated history–state pair under stop-gradient and trains continuation using recorded next-segment targets. Robot poses illustrate the method and are not hardware results.

Future music states predicted from arrived audio provide temporally aligned conditions for discrete structural planning and continuous-detail generation.

- **Kinematics-guided robot-native D+C generation.** Complementary reconstruction objectives assign explicit structural responsibilities to discrete codes, whose embeddings combine with continuous residuals through a shared boundary-conditioned decoder.
- **Commit Forcing for streaming continuation.** Joint commitment of generated structure and detail constructs a consistent history–state pair, enabling continuation training from the sampled reference transition.

On the evaluated 60 s FineDance generations, a frozen music–motion evaluator favors the intended soundtrack over alternative tracks in approximately 75% of pairwise comparisons under the reported evaluation and selection protocol. A separate 18-track study characterizes the effects of history regularization on motion fidelity, diversity, and smoothness. These findings describe musical correspondence and motion properties at the reference-generation level. Section IV clarifies their scope and the comparisons required to attribute benefits to musical forecasting, D+C supervision, and Commit Forcing, as well as to assess musical performance after execution.

sectionRelated Work

A. *Learning-Based Humanoid Control and Motion Tracking*

Reinforcement learning and motion imitation provide foundations for expressive whole-body control. Neural Categorical Priors [10] and Generative Pretrained Controllers [11] connect learned motion vocabularies with physics-based character control. Reference-conditioned policies further enable reusable motion tracking: PHC [12] demonstrates tracking and recovery in simulated avatars, while ExBody2 [13], BeyondMimic [14], and SONIC [9] advance expressive and versatile humanoid control. These methods establish how reference movement can be realized under physical dynamics. Our work uses a pretrained whole-body controller as the execution layer and focuses on generating the music-conditioned references it follows. The controller retains measured-state feedback, while the generator maintains its own committed reference context.

B. Music-Conditioned Motion Generation

Music-to-dance systems use explicit acoustic descriptors, such as those extracted with librosa [15], or learned audio representations from models including Jukebox [16], MERT [17], and MuQ [18]. AIST++/FACT [1] and Bailando [19] develop music-conditioned choreography through sequence modeling and discrete motion representations. EDGE [2] and LODGE [20] advance diffusion-based and long-form dance generation, while FineDance [3], MEGADance [21], and SoulNet [4] address fine-grained motion, genre-aware generation, and music–motion alignment.

Streaming generation additionally constrains the audio available before each motion decision. DiscoForcing [5] uses causal audio conditioning and diffusion forcing for streaming motion and humanoid deployment. Our music interface supplies explicit forecasts of upcoming music states from the observed prefix. Generative music models, including Live Music Models [22] and Magenta RealTime 2 [7], provide the continuation prior used for this purpose. We investigate how their predicted states can condition upcoming robot motion without access to future waveform samples. The music forecaster remains frozen; paired dance training learns how to use its predictions.

C. Motion Representations and Generation

Discrete motion models such as T2M-GPT [23] and MoMask [24] use quantized vocabularies, while MLD [25] generates motion through continuous latent diffusion. DisCoRD [26] connects discrete tokens with continuous decoding. Most closely, DC-Motion [8] combines discrete structure and continuous residuals. Our D+C design builds on this hybrid principle through robot-kinematic reconstruction objectives that assign designated structural responsibilities to discrete codes, with full motion reconstructed through a shared boundary-conditioned decoder.

Recursive generation also introduces a mismatch between recorded training context and generated inference context. Scheduled Sampling [27] and Diffusion Forcing [28] address related sequence-training problems. MotionStreamer [29] combines causal motion compression with diffusion-based autoregression and Two-Forward training. Commit Forcing focuses on the transition induced by our D+C generator: sampled structure and detail enter the committed history together, and their decoded motion determines the next reference boundary state. Its distinguishing operation is the joint construction of this history–state pair for continuation training.

D. Integrating Motion Generation with Humanoid Control

Connecting conditional generation to physical control enables humanoids to express higher-level intent. HARMON [30] generates whole-body humanoid motion from language descriptions. RoboPerform [6] combines motion content with audio-derived style through a teacher–student control framework, while DiscoForcing [5] connects streaming motion generation to humanoid execution through retargeting and tracking. These systems establish complementary interfaces between external conditions, generated movement, and robot control.

Other approaches use physical execution to inform generation or adapt the controller. PhysDiff [31] incorporates physics-based correction into denoising, and PARC [32] uses physically tracked motion to augment generator training. BeyondMimic [14] extends tracking to guided generative motion composition, while terrain-aware humanoid generation [33] couples robot-native references with tracker adaptation. Our framework performs retargeting during data preparation, generates robot-native references online, and delivers committed motion to fixed SONIC control. Continuation training operates within the generated reference process. We therefore distinguish reference generation, delivery, and measured execution, and assess musical correspondence and motion quality alongside tracking accuracy.

II. Method

ForeDance couples three decisions: what music to anticipate, how to express the response as structured motion, and how to continue after committing it. Figure 1 shows the complete system; the following figures expand the reconstruction and continuation mechanisms.

a) Streaming task: At decision time τ, $a_{\leq\tau}$ is the heard audio, h is committed code history, and b is the pose and velocity state reconstructed from the last committed reference frames. A frozen music model predicts $m^{\text{pred}} = F(a_{\leq\tau})$, and the generator samples

$$(q_{1:H}, r_{1:H}) \sim p_\theta(q, r \mid h, b, m^{\text{pred}}). \tag{1}$$

Here q is discrete structure and r is continuous detail. Their sum decodes to native G1 motion. Both streams commit the same first $C < H$ steps; the remaining plan is discarded. We use $H = 8$ and $C = 4$ at 15 latent steps per second, yielding 16 planned and eight committed frames at 30 Hz.

A. Anticipatory Music Conditioning

SpectroStream tokenizes heard audio, and frozen MRT-2 Small rolls forward to predict musical states [7]. Its music pretraining supplies the continuation prior; paired dance training learns how to turn that prior into movement. Fourteen forecast states at 25 Hz cover approximately 0.56 s, alongside the 0.53 s motion plan. Future waveform samples are never used as input.

a) Align time before modulating features: Forecast and motion steps have different rates, so matching array indices would assign the wrong musical time. For state u_j with physical lead ℓ_j relative to the decision, we form

$$c_j = W\,\mathrm{LN}(u_j) + \phi(\ell_j). \tag{2}$$

Here W projects the music features and ϕ encodes their lead time. We linearly interpolate neighboring projected states at motion lead $d_i = i/15$ to obtain $\hat{c}_i$. A coverage mask m_i is one only within the valid forecast support and zero when the condition is absent or no longer covers that motion time.

Each conditioned block adds a gated, frame-wise modulation [34]:

$$v_i' = v_i + m_i\, g \odot \left[s(\hat{c}_i) \odot \mathrm{LN}(v_i) + t(\hat{c}_i) \right]. \tag{3}$$

The learned maps s,t predict scale and shift, and g is a learned channel gate. When $m_i = 0$, the block retains its original activation exactly. Separate modulators condition the structural Transformer blocks and residual diffusion heads using the same physical-time alignment. The generator and these mappings are trained; MRT-2 remains frozen.

B. Structure–Detail Representation and Paired Training

An encoder maps native motion and its starting state to $z = E_\varphi(x,b)$. Quantization selects $q = Q(z)$ with codebook embedding $e(q)$, and a continuous residual retains the remaining information:

$$z = e(q) + r. \tag{4}$$

Both components are decoded by one boundary-conditioned decoder $D_\psi(\cdot, b)$. Their roles are learned through complementary reconstruction objectives, rather than separate anatomical decoders.

a) Learning complementary roles: We train two reconstructions with the same decoder and starting state:

$$x_f = D_\psi(e(q) + r, b), \qquad x_s = D_\psi(e(q), b). \tag{5}$$

The full path preserves complete movement, while the structure-only path gives the discrete anchor an explicit kinematic responsibility (Fig. 2). Codebook and commitment losses train the quantized representation; Appendix **??** specifies the gradient routes.

The full loss supervises native coordinates, native velocities, forward-kinematics positions, and geometry velocities. Each block is standardized with training-set statistics and receives equal weight. The structural loss uses the corresponding selected coordinates: root, legs, and waist in the native views, and torso, wrists, and soles in the geometry views. Consequently, wrist placement can be structural while arm-joint articulation remains part of the complementary native view. Auxiliary structural reconstruction under residual corruption and a boundary-transition loss reinforce these objectives; their settings and exact partitions remain in Appendix **??**.

b) Coupled motion generation: A causal Transformer samples q. Its sampled IDs are embedded and added to the residual model's input features; a diffusion generator [35] samples r using this structure, motion history, boundary state, and aligned music. The codebook embedding and sampled residual then sum at the shared decoder to produce 38-dimensional native motion. Joint commitment preserves the pairing of the two streams across planning steps.

C. Commit Forcing

Teacher Forcing supplies recorded history and a recorded starting state; streaming inference supplies both from the model's earlier output. Commit Forcing changes this conditioning transition while retaining recorded supervision for the next segment (Fig. 3).

a) Construct the next history and state together: A first pass samples $\tilde{q}, \tilde{r}$ with the inference samplers under stop-gradient. Their common committed segment updates the two parts of the next context:

$$(\tilde{h}^+, \tilde{b}^+) = U(h, b, \tilde{q}_{1:C}, \tilde{r}_{1:C}). \tag{6}$$

U appends the committed codes to history and reconstructs the boundary state from their decoded reference motion. Context replacement selects this entire pair together, so generated history is accompanied by the state of the same commitment. Each supplied training example constructs one such transition independently; the first pass is not a recursive long rollout.

b) Keep the target, change its residual coordinates: The next recorded structural target q^* and full latent $z^* = e(q^*) + r^*$ remain fixed. The next structure model is supervised against q^*, while its sampled structure $\hat{q}$ conditions residual denoising. We express the residual target relative to that sampled embedding:

$$r^*_{\text{rebased}} = z^* - e(\hat{q}), \qquad e(\hat{q}) + r^*_{\text{rebased}} = z^*. \tag{7}$$

Thus the structure and residual losses retain a common recorded target latent despite a changed structural sample. This is latent rebasing, not target re-encoding under $\tilde{b}^+$: the decoder also depends on its boundary state, so latent equality does not imply identical decoded motion.

c) History regularization: After context construction, full-history corruption (FHC) perturbs discrete embeddings and residuals across the motion history. It preserves boundary state, music, masks, and timestamps, and is disabled during inference. Appendix **??** specifies this separate training regularizer.

D. Robot Reference Integration and Execution

Figure 4 separates reference delivery from controller feedback. The committed native motion provides the reference interface to a separate, pretrained SONIC whole-body controller [9]. A reference bridge reconstructs root motion in the fixed sequence coordinate system, maps joint references into the controller convention, and resamples the 30 Hz motion onto a continuous 50 Hz grid. Appendix **??** defines the coordinates and state reconstruction. Only committed motion is delivered; the uncommitted suffix is discarded before the next plan and is not sent to the controller.

SONIC remains fixed and combines the consumed reference with measured robot state to produce control actions. The generator separately updates its latent history and boundary state from committed reference motion. Appendix **??** specifies the delivery and measurement protocol.

III. Experiments

We separate evaluation of the music–motion evaluator on reference motion from its use to assess generated dance. Beat alignment and motion-quality measures complement learned correspondence. Supporting history-training comparisons examine motion quality; the core ablations still awaiting results are marked explicitly in Section III-D.

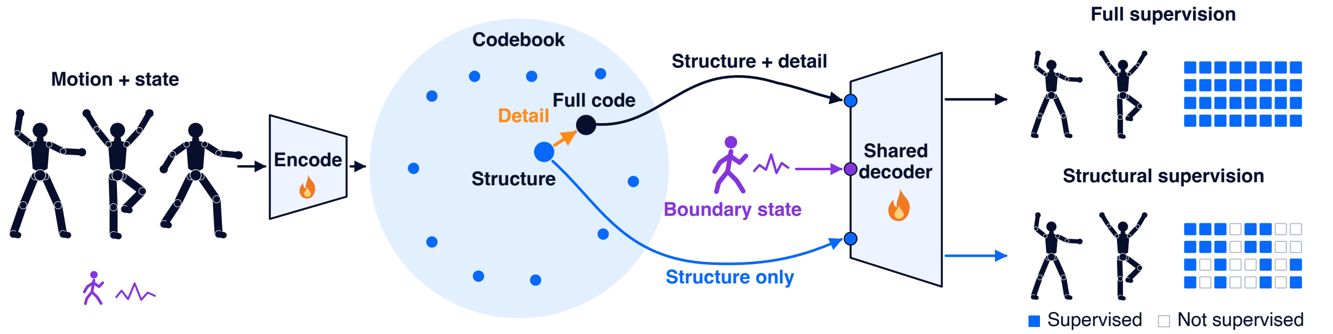


Fig. 2. **Learning complementary structure and detail.** The codebook anchor and continuous offset form the full code. Both paths share one state-conditioned decoder and produce complete motions. Filled cells select coordinates for supervision in four blocks: native motion, native velocity, Cartesian position, and Cartesian velocity. Counts are schematic; flames mark learned modules.

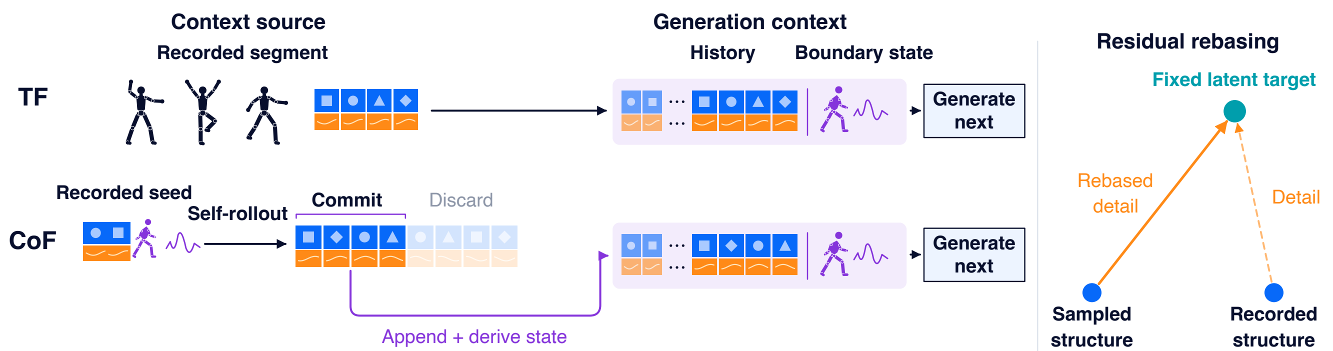


Fig. 3. **Constructing generation contexts with Commit Forcing.** TF uses recorded history and boundary state. CoF first samples a plan from a recorded seed context, appends its committed prefix to history, and derives the paired boundary state from the decoded committed motion. The discarded tail does not enter the next context. At right, residual rebasing retains the recorded target latent while changing the structural origin. Tile counts and geometry are schematic.

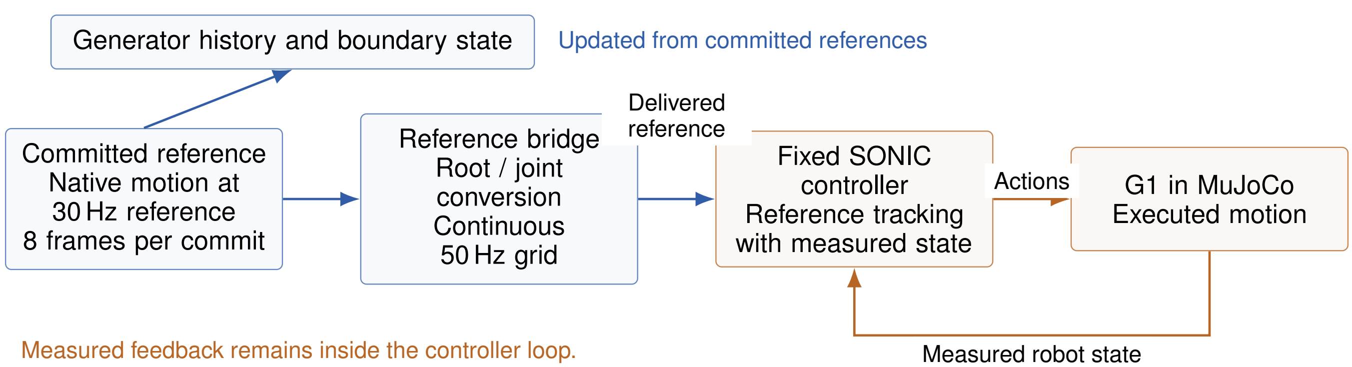


Fig. 4. **From committed references to motion tracking.** The bridge converts the reference stream onto the controller grid. Committed references update the generator context, whereas measured robot state closes the SONIC tracking loop.

A. Setup and Metrics

Evaluator validation uses reference motion from 30 content groups, comprising 3,661 windows and 15 disjoint song pairs per discrimination task. Generator correspondence uses 99 generated 60 s FineDance trajectories: eleven tracks, three independently trained generators, and three sampling seeds per generator. The separate beat-alignment and motion-quality study uses all 18 archived tracks, a common 23 s segment, and the same three-by-three training/sampling repeat structure, giving 162 outputs per configuration. All motion is represented in native G1 space. Appendices **??** and **??** specify the training and cohort protocols.

TABLE I

EVALUATOR VALIDATION ON REFERENCE MOTION. THE FROZEN EVALUATOR RANKS INTENDED MUSIC AGAINST ORDINARY OR MUSICALLY SIMILAR ALTERNATIVES. PAIRWISE ACCURACY IS IN PERCENT; RANDOM ORDERING IS 50%.

Discrimination task	Accuracy (%)
Ordinary alternative music	68.41
Similar alternative music	64.08

TABLE II

MUSIC ALIGNMENT OF FOREDANCE GENERATIONS. LEARNED CORRESPONDENCE IS PAIRWISE SOUNDTRACK ACCURACY; BAS MEASURES MOTION-TO-MUSIC BEAT ALIGNMENT. HIGHER IS FAVORABLE FOR BOTH METRICS; THEIR EVALUATION COHORTS AND SCORE SCALES DIFFER.

Measure	Evaluation cohort	Score
Learned correspondence	11 tracks, 60 s	75.01%
Beat alignment (BAS)	18 tracks, 23 s	0.4419

a) Learned music–motion correspondence: Motivated by music–motion retrieval [4], we use a frozen G1-native evaluator trained independently of the generator. Its score $s_w(x,a)$ averages cosine similarities from three evaluator instances for five-second music–motion windows. For intended music a_i and an alternative track a_j, the correspondence margin is

$$\delta_{ijw} = s_w(x_i,a_i) - s_w(x_i,a_j). \tag{8}$$

Pairwise accuracy counts positive margins, with half credit for ties; random ordering gives 50%. The evaluator-validation tasks use ordinary or musically similar alternative tracks. Generator scoring compares each fixed motion against its intended music and all ten alternatives in its fixed gallery, preserving native audio offsets. Windows and sampling repeats are aggregated within training-seed/query-track cells. Appendix **??** defines the frozen evaluator.

b) Beat alignment and motion quality: Beat Alignment Score (BAS) measures the proximity of motion beats to music beats. We use local minima of smoothed mean forward-kinematics keypoint speed and a motion-to-music Gaussian matching score with a 0.10 s scale; Appendix **??** gives the exact definition. BAS measures rhythmic timing, while learned correspondence assesses soundtrack preference. Kinetic and geometric Fréchet distances (FID_k, FID_g) compare generated and reference distributions of motion dynamics and body configurations. Div_k and Div_g measure spread in the same feature spaces. PFC is a contact-related physical-consistency proxy, complemented by root/contact and jerk diagnostics. The 23 s and 60 s cohorts are analyzed separately.

B. Evaluator Validation

Table I evaluates the score itself on reference motion. Pairwise accuracy is 68.41% against ordinary alternative music and 64.08% against musically similar alternatives. Both exceed random ordering in point estimate, with the similar-music task providing a more demanding correspondence check. These results characterize the evaluator's ability to distinguish soundtracks, rather than the quality of ForeDance generations.

C. Generator Evaluation

Table II reports learned soundtrack correspondence and beat alignment for generated motion. The intended soundtrack wins 75.01% of pairwise comparisons, with a mean correct-minus-alternative score margin of 0.1783. Every generator training seed has a positive mean margin. The separate 18-track study yields BAS 0.4419. These complementary measures characterize soundtrack preference and rhythmic alignment; their different cohorts preclude treating the values as a single pooled evaluation. Appendix **??** retains the detailed score statistics and descriptive intervals.

a) History-training comparison: Table III compares clean history, history dropout, and full-history corruption (FHC), holding the remaining settings fixed. Relative to clean history, FHC reduces kinetic/geometric FID by 21.9%/15.0%, with both directions favorable in all three training seeds, and increases kinetic diversity. BAS rises from 0.4352 to 0.4419; this is a descriptive change, not a significance claim. PFC and jerk favor clean history, so the distribution gains coexist with a smoothness tradeoff. This comparison examines history regularization, rather than isolating the benefit of musical anticipation or Commit Forcing.

b) Qualitative generation: Figure 5 shows reference poses at 10, 30, and 50 s from one selected 60 s rollout. These original frames illustrate body configuration across the sequence; they do not measure temporal smoothness.

D. Pending Ablation Studies

TODO for this working draft: the following controlled studies await results and are not counted as completed evidence.

a) TODO: musical anticipation: Compare forecast-conditioned and heard-music-conditioned generators with matched training, tracks, and sampling. Report learned correspondence, BAS, and motion quality to isolate the value of predictive musical context.

b) TODO: paired representation training: Complete current-representation reconstruction and intervention results to test whether structure and detail serve their intended complementary roles. Include non-collapse checks and matched supervision comparisons.

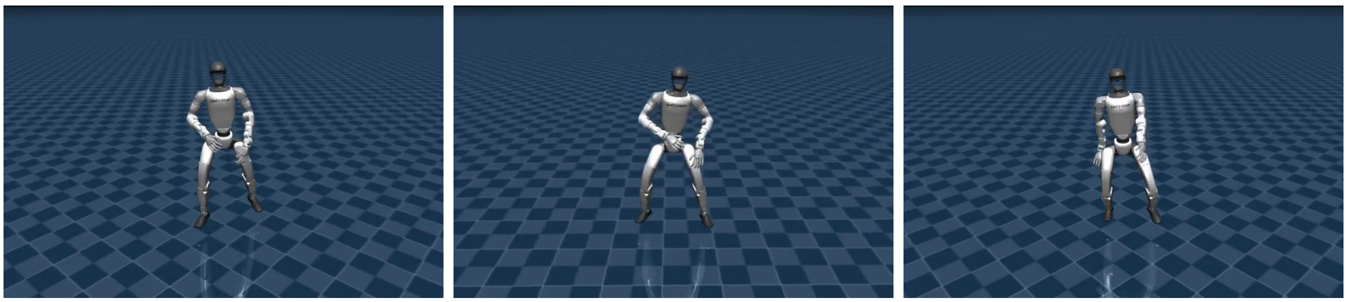
c) TODO: single-representation controls: Compare independently trained discrete-only, continuous-only, and combined structure–detail generators under matched settings. Removing a decoder branch does not substitute for a separately trained baseline.

d) TODO: committed continuation: Compare Commit Forcing with Teacher Forcing on matched 60 s free-running sequences. Report aggregate quality and each of the three non-overlapping 20 s windows, including commitment-boundary behavior and matched qualitative sequences.

TABLE III

Motion quality and beat alignment under matched history training. All three configurations share initialization, architecture, music conditioning, the complete 18-track cohort, common 23-second segments, three training seeds, and three sampling seeds. Lower is favorable for FID and PFC; higher BAS indicates closer beat alignment. Diversity describes motion spread and is interpreted together with fidelity.

History training	FID_k	FID_g	Div_k	Div_g	BAS	PFC
Clean history	208.599	0.482	7.495	0.493	0.4352	0.143
History dropout	164.761	0.469	7.801	0.473	0.4338	0.182
History corruption	162.831	0.410	8.626	0.508	0.4419	0.167



10 s 30 s 50 s

Fig. 5. **A 60-second ForeDance generation.** Original frames at 10, 30, and 50 s from a selected FineDance sequence. The images show rendered reference motion.

E. Supporting Execution Diagnostics

The interface in Section II-D separates the generated, resampled, controller-consumed and measured trajectories. The supporting study replays four saved trajectories from one FineDance track through fixed SONIC in MuJoCo. These trajectories use an earlier 34-dimensional representation, separate from the current 38-dimensional generation cohort; audio inference is not run during replay.

After excluding the first two seconds and aligning joint trajectories, median amplitude retention ranges from 0.858 to 0.901, and median squared-velocity retention from 0.411 to 0.496. Position tracking therefore coexists with dynamic attenuation. These diagnostics characterize the reference-to-execution path; they do not establish current-model hardware performance or comparative musical benefit. Appendix **??** retains the complete measurements and protocol.

IV. Discussion

a) Supported findings and attribution: The generator studies characterize soundtrack correspondence, beat alignment, and a distribution–smoothness tradeoff in history regularization. The evaluator's reference-motion validation is a separate measurement. Attributing gains to anticipation, representation design, or committed continuation requires the controlled studies listed in Section III-D. Broader music pretraining supplies a transferable prediction prior, while improved out-of-distribution dance generalization remains a hypothesis. The historical replay diagnostics characterize tracking separately from these current reference-generation results.

b) Evaluation scope and selection: The evaluator configuration was selected after inspection of its 30-content-group calibration cohort. For compact reporting, generator music scores use the higher complete initialization result for each metric after evaluation; all tracks and sampling repeats in the respective cohort are retained. The history-training comparison uses one shared initialization. Calibration scores and retained intervals are descriptive, not selection-independent evidence of superiority. The score has not been validated against human preference or executed trajectories. The motion-quality cohort is separate, and the illustrated sequence is a selected qualitative case. Its complete source and selection record are retained with the evidence.

V. Conclusion

ForeDance brings anticipatory music conditioning to structured streaming dance: a frozen music foundation model predicts prospective context, kinematics-guided structure–detail training organizes motion components, and Commit Forcing learns continuation after their joint commitment. Completed studies find consistent soundtrack correspondence and characterize motion quality under matched history-training choices. Together, the method and reference interface provide an explicit framework for studying how musical foresight translates into continuously generated robot motion.

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